Biswajyoti Nath

Computer Science Engineer | Cybersecurity & Quantum Cryptography

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SUMMARY

Computer Science Engineering student specializing in cybersecurity and emerging quantum-safe technologies.
Experience in Python-based ML systems, secure protocol concepts, and applied computer vision. Published analytical work on randomness in quantum cryptography and building foundations in post-quantum cryptographic systems and distributed security models.

EDUCATION

Bachelor of Technology - Computer Science & Engineering, *Barak Valley Engineering College (ASTU)* 08/2024 – Present | Sribhumi, Assam

- Awarded the Ishan Uday Scholarship (Government of India) for academic merit, providing full financial support during undergraduate studies.

High School Diploma, *Jawahar Navodaya Vidyalaya* 09/2022 – 05/2024 | Rangia, Assam

SKILLS

Programming
- Python
- C , C++
- Object-Oriented Programming (OOP)
- Data Structures and Algorithms

Tools
- Git & GitHub
- Ubuntu
- VS Code

Cybersecurity
- Linux Security
- Network Analysis
- Cryptography Fundamentals
- Ethical Hacking

Conceptual / Technical Skills
- Analytical Skills
- Software Development Best Practices

CERTIFICATIONS

Python Essentials 1 - Cisco Networking Academy | Python Institute

Python Essentials 2 - Cisco Networking Academy | Python Institute

Ethical Hacker - Cisco Networking Academy: Trained in vulnerability assessment and ethical security testing practices.

IBM Quantum Enigmas (Levels I- II): Completed advanced quantum logic challenges using circuit abstractions, controlled operations, and measurement- based state elimination.

Research and Publications

- Comparative analysis of pseudorandom vs quantum random number generators in quantum key distribution protocols
- Implemented reproducible experiments using Qiskit demonstrating QRNG advantages in secure communication
- Received conceptual clarification from Prof. Gilles Brassard and Prof. Nicolas Gisin regarding protocol assumptions and randomness in quantum cryptography.
- Published with open-access DOI for academic transparency and reproducibility.

INVOLVEMENT

Rotary Club Silchar , *Volunteer* 07/2025 – 08/2025 | Silchar
- Participated in organizing health campaigns and social welfare events.
- Conducted STEM teaching sessions for students in underserved communities.

Workshop Participant 03/2025 – 03/2025
Encoders Pro Pvt. Ltd.
- Hands-on exposure to core cybersecurity concepts, threat models, and basic security workflows.